Infosys System Engineer: Mock Test

Total questions: 60 | **Duration:** 120 minutes | **Sectional cut-offs apply**

Section	Questions	Time
1. Reasoning Ability	Q1–15	25 min
2. Technical Ability (Quant)	Q16–25	35 min
3. Verbal Ability	Q26–45	20 min
4. Pseudo Code	Q46–50	10 min
5. Numeric Puzzle Solving	Q51–54	10 min
6. English Grammar	Q55–59	10 min
7. English Writing	Q60	10 min

SECTION 1: Reasoning Ability (15 Q, 25 min)

Q1. Statements: All engineers are coders. Some coders are testers. No tester is a manager.
Conclusions: I. Some coders are not managers. II. Some engineers are testers. III. No engineer is a manager.
(a) Only I follows (b) I and II follow (c) I and III follow (d) All follow

Q2. Statements: Some pens are books. All books are chairs. No chair is a table.
Conclusions: I. Some pens are chairs. II. No book is a table. III. Some pens are not tables. IV. Some tables are pens.
(a) I and II only (b) I, II and III only (c) II, III and IV only (d) All follow

Q3. *Data Sufficiency.* For integers x and y, is $x > y$?
I. $x^2 > y^2$. II. $x + y > 0$.
(a) I alone is sufficient (b) II alone is sufficient (c) Both together are sufficient, neither alone (d) Even both together are not sufficient

Q4. *Data Sufficiency.* What is Riya's present age?
I. Riya is twice as old as her sister, who is 3 years younger than their brother.
II. Five years hence, the sum of the ages of Riya and her sister will be 40.
(a) I alone is sufficient (b) II alone is sufficient (c) Both together are sufficient, neither alone (d) Even both together are not sufficient

Q5. In a certain code, MONDAY is written as NNOCBX. How is SUNDAY written in that code?
(a) TTOCBX (b) TVOEBZ (c) TTOCBZ (d) RTOCBX

Q6. In a code, "sun rises early" = "ka mo ti", "early bird catches" = "ti ra po", and "bird sun light" = "ra ka du". What is the code for "rises bird light"?
(a) mo ra du (b) mo po du (c) ka ra du (d) mo ti du

Q7. A solid cube painted red on all faces is cut into 64 identical small cubes ($4 \times 4 \times 4$). How many small cubes have exactly two faces painted?
(a) 12 (b) 24 (c) 36 (d) 8

Q8. In triangle ABC, the medians AD, BE and CF are drawn and meet at G. How many triangles are there in the figure in total?

(a) 12 (b) 14 (c) 16 (d) 18

Q9. Pointing to a man, Meera said, "His mother is the only daughter of my mother." How is Meera related to the man?

(a) Mother (b) Sister (c) Aunt (d) Grandmother

Q10. If A + B means A is the mother of B, A − B means A is the brother of B, and A × B means A is the father of B, which expression shows that P is the maternal uncle of R?

(a) P + Q − R (b) P − Q + R (c) P × Q − R (d) P − Q × R

Q11. Ravi walks 10 m north, turns right and walks 6 m, turns right and walks 14 m, turns left and walks 3 m, then turns left and walks 4 m. Where is he now relative to the starting point?

(a) 9 m East (b) 9 m West (c) 11 m South-East (d) 5 m North

Q12. A person facing North-East turns 90° clockwise, then 135° anticlockwise, then 180° clockwise. Which direction is he facing now?

(a) North (b) South (c) South-West (d) East

Q13. Find the next term: 3, 7, 16, 35, 74, ?

(a) 143 (b) 153 (c) 149 (d) 158

Q14. Find the next term: 4, 9, 25, 49, 121, 169, ?

(a) 289 (b) 259 (c) 361 (d) 323

Q15. Find the next letter: B, E, I, N, T, ?

(a) A (b) Z (c) B (d) Y

SECTION 2: Technical Ability, Quantitative (10 Q, 35 min)

Q16. In how many ways can the letters of COMMITTEE be arranged so that no two vowels are adjacent?

(a) 5400 (b) 3600 (c) 10800 (d) 2700

Q17. A committee of 5 is to be formed from 5 boys and 4 girls with at least 2 boys and at least 2 girls. In how many ways can this be done?

(a) 90 (b) 100 (c) 105 (d) 120

Q18. Two fair dice are thrown. Given that at least one die shows 4, what is the probability that the sum is a prime number?

(a) 4/11 (b) 1/3 (c) 5/11 (d) 4/9

Q19. A bag has 5 red, 4 blue and 3 green balls. Three balls are drawn at random without replacement. What is the probability that all three are of different colours?

(a) 3/11 (b) 1/4 (c) 2/11 (d) 5/22

Q20. A shopkeeper marks his goods 40% above cost price and allows two successive discounts of 10% and 15%. His net profit percentage is:

(a) 7.1% (b) 5.5% (c) 8.6% (d) 6.4%

Q21. A, B and C can individually finish a job in 12, 18 and 24 days respectively. They work on alternate days in the order A, B, C, A, B, C, … (one person per day, starting with A). On which day is the job completed?
(a) 16th (b) 17th (c) 18th (d) 19th

Q22. Two trains of lengths 240 m and 160 m run on parallel tracks. Running in opposite directions they cross each other in 8 s; running in the same direction the faster overtakes the slower in 40 s. The speed of the faster train is:
(a) 90 km/h (b) 108 km/h (c) 120 km/h (d) 72 km/h

Q23. In an election, 20,000 voters were registered. 80% voted, and 5% of the votes cast were invalid. The winner got 55% of the valid votes. By how many votes did he win?
(a) 1520 (b) 1600 (c) 1360 (d) 1440

Q24. The difference between compound interest (annual) and simple interest on a sum for 3 years at 10% p.a. is ₹310. Find the sum.
(a) ₹8,000 (b) ₹9,000 (c) ₹10,000 (d) ₹12,000

Q25. ₹12,000 is invested partly at 8% simple interest and the rest at 10% compounded annually. After 2 years the total interest is ₹2,270. How much was invested at simple interest?
(a) ₹4,000 (b) ₹5,000 (c) ₹6,000 (d) ₹7,000

SECTION 3: Verbal Ability (20 Q, 20 min)

Read the passage and answer Q26–30.

> *In software organisations, "technical debt" is often treated as a purely engineering concern, a messy residue of hurried decisions that developers will eventually tidy up. This framing is misleading. Debt, in the financial sense, is not inherently harmful; a company that borrows to seize a market opportunity may be acting shrewdly. The problem arises when the interest, meaning the extra effort every future change demands, is never acknowledged on the balance sheet. Teams then find that features which once took days now take weeks, yet managers, seeing only the delivered functionality, attribute the slowdown to declining developer productivity. Proponents of disciplined refactoring argue that a fixed share of every sprint should be reserved for repayment. Critics counter that such quotas are arbitrary and encourage cosmetic rewrites that leave real risks untouched. A more defensible approach, some suggest, is to make debt visible by tracking how often particular modules cause delays or defects, and to repay only where the interest is demonstrably high.*

Q26. The central idea of the passage is that:
(a) Technical debt is harmful and must be eliminated at once
(b) Technical debt is not necessarily bad, but its recurring costs must be made visible and repaid selectively
(c) Managers, not developers, are responsible for technical debt
(d) Fixed refactoring quotas are the best way to manage debt

Q27. According to the passage, managers blame developer productivity because they:
(a) do not trust developers
(b) see only the delivered features and not the hidden extra effort
(c) are unaware that software can be refactored
(d) have reserved too little time for sprints

Q28. Which statement would critics of fixed repayment quotas most likely agree with?
(a) Repayment effort should target genuine risks rather than meet a set proportion
(b) Refactoring should be abandoned entirely
(c) Cosmetic rewrites are the most valuable form of repayment
(d) Sprints should be lengthened

Q29. In the passage, "interest" most nearly refers to:
(a) money paid to lenders
(b) the recurring additional effort imposed on future changes
(c) the curiosity of developers
(d) the profit from a market opportunity

Q30. Which of the following is NOT supported by the passage?
(a) Borrowing can sometimes be a shrewd decision
(b) A slowdown in delivery always indicates falling developer productivity
(c) Tracking defect-prone modules can help identify costly debt
(d) Debt is often left unrecorded in practice

Sentence Correction (Q31–35). Choose the best replacement for the bold part.

Q31. Neither the manager nor the **engineers was** aware of the revised deadline.
(a) No improvement (b) engineers were (c) engineers has been (d) engineer were

Q32. Had the team would have tested the patch thoroughly, the outage could have been avoided.
(a) No improvement (b) Had the team tested (c) Has the team tested (d) If the team would test

Q33. Hardly **the server had restarted when** the alerts began flooding in.
(a) No improvement (b) had the server restarted when (c) did the server restarted when (d) the server restarted than

Q34. The number of applicants **are** increasing, but a large number of them **lacks** the required skills.
(a) are … lacks (b) is … lack (c) is … lacks (d) are … lack

Q35. Each of the interns, along with their mentor, **were** invited to present.
(a) No improvement (b) was (c) are (d) have been

Para-jumbles (Q36–38). Arrange the sentences in the correct order.

Q36.
P: However, the prototype failed under real-world load.
Q: The engineers were confident that their design was flawless.
R: Post-mortem analysis revealed that no load testing had been performed.
S: Consequently, the launch was postponed by three months.
(a) QPRS (b) QRPS (c) PQRS (d) QSPR

Q37.

P: Rather, they encourage people to think about how the story might have unfolded differently.

Q: Counterfactual questions such as "What if the treaty had never been signed?" are popular with historians.

R: They do not aim to predict what would actually have happened.

S: Such speculation, when disciplined, can illuminate the forces that shaped real events.

(a) QRPS (b) QPRS (c) RQPS (d) QRSP

Q38.

P: The first industrial robots were deployed on car assembly lines in the 1960s.

Q: Today, collaborative robots work safely alongside humans.

R: These early machines were caged off, since they could not sense human presence.

S: Advances in sensors and software have since removed this limitation.

(a) PRSQ (b) PQRS (c) PSRQ (d) RPSQ

Vocabulary, Synonyms and Antonyms (Q39–45).

Q39. The auditor's **meticulous** approach uncovered discrepancies others had missed. "Meticulous" means:

(a) hasty (b) painstakingly careful (c) aggressive (d) casual

Q40. The contract was deliberately worded to **obfuscate** the penalty clauses. "Obfuscate" means:

(a) highlight (b) simplify (c) make obscure (d) cancel

Q41. Fill in the blank: The startup's success proved _____; within a year it had vanished.

(a) enduring (b) ephemeral (c) robust (d) inevitable

Q42. Synonym of **UBIQUITOUS**:

(a) rare (b) omnipresent (c) ancient (d) hidden

Q43. Synonym of **RECALCITRANT**:

(a) obedient (b) obstinate (c) cheerful (d) timid

Q44. Antonym of **LOQUACIOUS**:

(a) talkative (b) taciturn (c) eloquent (d) boisterous

Q45. Antonym of **MAGNANIMOUS**:

(a) generous (b) noble (c) petty (d) tolerant

SECTION 4: Pseudo Code (5 Q, 10 min)

Q46. What is printed?

```
int f(int n) {
    if (n <= 1) return n;
    return f(n-1) + 2 * f(n-2);
}
print(f(6))
```

(a) 15 (b) 21 (c) 27 (d) 31

Q47. What is printed?

```
x = 45; count = 0;
while (x != 0) {
    x = x & (x - 1);
    count++;
}
print(count ^ (45 >> 2))
```

(a) 4 (b) 11 (c) 15 (d) 7

Q48. What is printed?

```
s = 0
for i = 1 to 5
    for j = i to 5
        if ((i + j) % 3 == 0) continue
        if (j == 5) break
        s = s + i * j
    end for
end for
print(s)
```

(a) 38 (b) 42 (c) 46 (d) 50

Q49. A stack is used (operands are evaluated left to right). What is printed?

```
push(3); push(5); push(7);
x = pop();
push(x * 2);
push(pop() + pop());
print(pop() - pop())
```

(a) 14 (b) 16 (c) 19 (d) 22

Q50. What is the final content of `arr`? (indexing starts at 0; updates are in place)

```
arr = [4, 1, 3, 2]
for i = 0 to 3
    arr[i] = arr[i] + arr[arr[i] % 4]
end for
```

(a) [8, 2, 5, 7] (b) [8, 2, 5, 6] (c) [8, 5, 5, 7] (d) [5, 2, 5, 7]

SECTION 5: Numeric Puzzle Solving (4 Q, 10 min)

Q51. *Matchstick.* A 2 × 2 grid of squares is made using 12 matchsticks (giving 4 small squares and 1 large square). What is the minimum number of sticks to remove so that no square of any size remains?

(a) 2 (b) 3 (c) 4 (d) 5

Q52. *Circular wheel.* Each wheel has numbers at Top, Right, Bottom, Left and a Centre number formed by the same rule.

Wheel 1: T = 3, R = 5, B = 4, L = 7, Centre = 14

Wheel 2: T = 2, R = 6, B = 5, L = 3, Centre = 7

Wheel 3: T = 4, R = 3, B = 6, L = 9, Centre = ?

(a) 21 (b) 27 (c) 30 (d) 33

Q53. *Matrix logic.* Find the missing number.

3	4	25
5	12	169
6	8	?

(a) 64 (b) 100 (c) 196 (d) 144

Q54. *Pattern.* Find the next term: 2, 12, 36, 80, 150, ?
(a) 216 (b) 252 (c) 240 (d) 294

SECTION 6: English Grammar (5 Q, 10 min)

Q55. Neither of the two proposals, despite several revisions, _____ approved.
(a) have been (b) has been (c) were (d) are

Q56. By the time the auditors arrive next month, we _____ the migration.
(a) completed (b) will complete (c) will have completed (d) have completed

Q57. The manager insisted _____ reviewing every line of code, despite being averse _____ delay.
(a) on, to (b) in, from (c) upon, with (d) at, to

Q58. *Error spotting.* Find the part with the error: "Not only did the CEO apologise (A) / but he also (B) / promised to (C) / compensated the customers. (D)"
(a) A (b) B (c) C (d) D

Q59. She is ___ European who works as ___ engineer at ___ MNC.
(a) a, an, an (b) an, an, a (c) a, an, a (d) a, a, an

SECTION 7: English Writing (1 Q, 10 min)

Q60. *Write an email of 120–150 words.*
You are a System Engineer in a project team. A production deployment scheduled for tonight has to be postponed by 48 hours because a critical defect was found during final testing. Write an email to the client project manager that informs them of the delay, explains the reason, states the corrective action and the revised timeline, and apologises for any inconvenience.

Evaluation rubric (10 marks): Subject line and format (2) | Clarity and completeness of content (3) | Professional tone and courtesy (2) | Grammar, spelling and vocabulary (2) | Word limit adherence (1)